

# Patient management



- **Not all dental radiographic techniques can be successfully performed on all patients. Radiographic examination techniques must often be modified to accommodate patients with special needs.**
  - **The dental radiographer must be competent in altering radiographic techniques to meet the specific diagnostic needs of individual patients.**
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**\*We discuss how to manage**

▮ **patients with a Gag reflex**

▮ **patients with physical or developmental disabilities**

▮ **pediatric patients**

▮ **edentulous patients.**



# 1. Gagging reflex

- **A gag reflex** is a protective mechanism of the body that serves to clear the airway of obstruction.
- In the dental radiography a hypersensitive gag reflex is **a common problem**.
- Intra oral films are placed in oral areas specifically related to the initiation of this reflex. Some patients may unconsciously activate this reflex **as a defense against anticipated unpleasantness**

■ The areas that are most likely to elicit the gag reflex when include the soft palate and the lateral posterior third stimulated of the tongue.

■ The precipitating factors that are responsible for initiating the gag reflex include (psychogenic stimuli) and (tactile stimuli)

**Prior to gag reflex, two reactions occur :**

- 1. Cessation of respiration.**
  - 2. Contraction of the muscles of the throat and abdomen.**
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# Patient management

- 1- Operator attitude :**
  - 2- patient and equipment preparations :**
  - 3- Exposure sequencing :**
  - 4- Film placement and technique :**
  - 5-use a topical anesthetic agent**
  - 6- extraoral radiograph :**
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## Operator attitude :

Most gagging can be controlled through the creation of the patient confidence or by-diverting the patient's concentration away from gagging reflex like asking the patient to breathe deeply through the nose or moving his arm.



## ≡patient and equipment preparations :

In a patient with hypersensitive gag reflex, every effort should be made to limit the amount of time that a film remains in the mouth. The longer a film stays in the mouth, the more likely the patient to gag.

## ≡ Exposure sequencing :

It play an important role in preventing gag reflex. The dental radiographer should always start with anterior exposure; with posterior exposure the dental radiographer should always expose the premolar before molar.

## ≡ Film placement and technique :

Each film must be placed and exposed as quickly as possible.

## ≡ Placement and technique modification include the following :

**a. Avoid the palate :** position the film lingual to the teeth and then firmly bring the film in to palatal tissue using one decisive motion.

**b. Demonstrate film placement:** in the areas that are most likely to elicit the gag reflex, rub a finger along the tissues near the intended area of film placement, while telling the patient, "This is where the film will be positioned." Then quickly place the film.



### **≡ use a topical anesthetic agent:**

when the cause of gag reflex is tactile , in such cases with gag reflex we can use local anesthesia. Salt application on the palate said to be useful.

### **≡ use of extraoral radiograph :**

If intra oral radiograph is impossible to obtain, so the dental radiographer must use extra oral films such as panoramic or lateral jaw radiographs to obtain.

## ***2. Physical Disabilities***

**1- Vision impairment :**

**2- Hearing impairment :**

**3- Mobility impairment :**

# Vision impairment

■ if a patient is blind or visually impaired, the dental radiographer must communicate using clear verbal explanations. The dental radiographer must keep the patient informed of what is being done and explain each procedure before performing it.

# Vision impairment

■ if a person is deaf or hearing impaired, the dental radiographer may ask the caretaker to act as interpreter, or use written instructions. When the patient can read lips, the dental radiographer must face the patient and speak clearly and slowly.



# Mobility impairment

- if a person is in a wheelchair and does not have use of the lower limbs, the dental radiographer may offer to assist the patient in transferring to the dental chair. If a transfer is not possible, the dental radiographer may attempt to perform the procedure with the patient seated in the wheelchair. If a person does not have use of the upper limbs and a holder cannot stabilize the film placement, the dental radiographer may ask the caretaker to assist with film holding

### ***3-Child patient :***

1. Use conveniently small intraoral films
  2. lead apron; the growing tissue of the child are particularly susceptible to the effects of ionizing radiation and must be protected, so lead apron and thyroid collar must be placed on the child prior to the x-ray exposure.
  3. Exposure factors (mA , kVp , time); must be reduced.
  4. Extraoral films can be used instead of intraoral.
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**5. Radiographic procedures should not be hurried nor should the child be apprised of any negative possibility like you might hurt a bit or you may gage.**

**6. Show him a radiograph of another child's teeth and let him see the radiographic procedure on another member of the family.**



# **Edentulous patient :**

**Edentulous patient require dental radiograph for the following reasons :**

- 1. Detect the presence of root tips, impacted teeth and lesions.**
  - 2. To establish the position of normal anatomic land marks relative to the crest of alveolar ridge .**
  - 3. To observe the quantity and quality of bone that is present.**
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**The radiographic examination of edentulous patient may include :**

**1- Panoramic radiograph :**

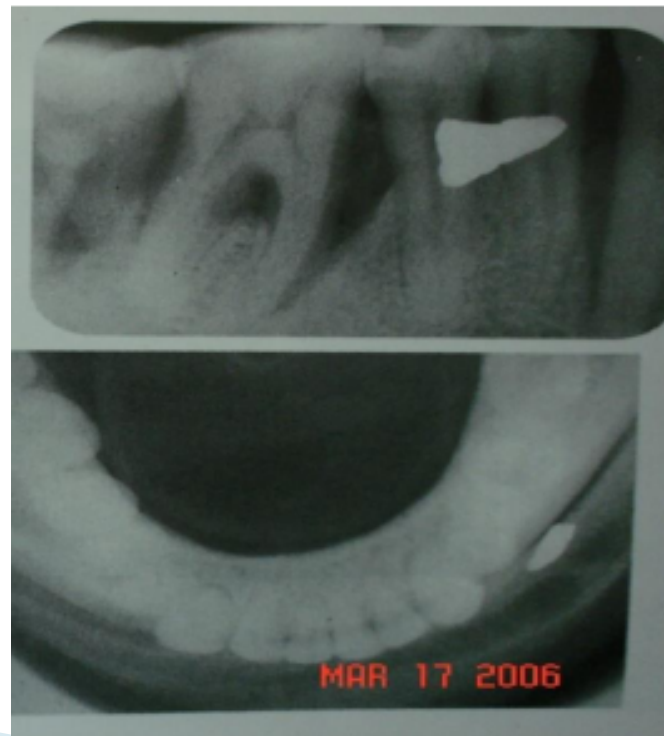
**2- Periapical radiograph :**

**3- Occlusal and periapical examination.**



# Localization techniques

- Is a method used to locate the position of a tooth or object in the jaws.
- The radiograph presents as two dimensional image of a three-dimensional object.



# Localization technique may be used to locate the followings:

- 1) foreign bodies
  - 2) impacted teeth
  - 3) unerrupted teeth
  - 4) retained roots
  - 5) salivary stones
  - 6) root positions
  - 7) jaw fractures
  - 8) broken needles & instruments
  - 9) filling materials
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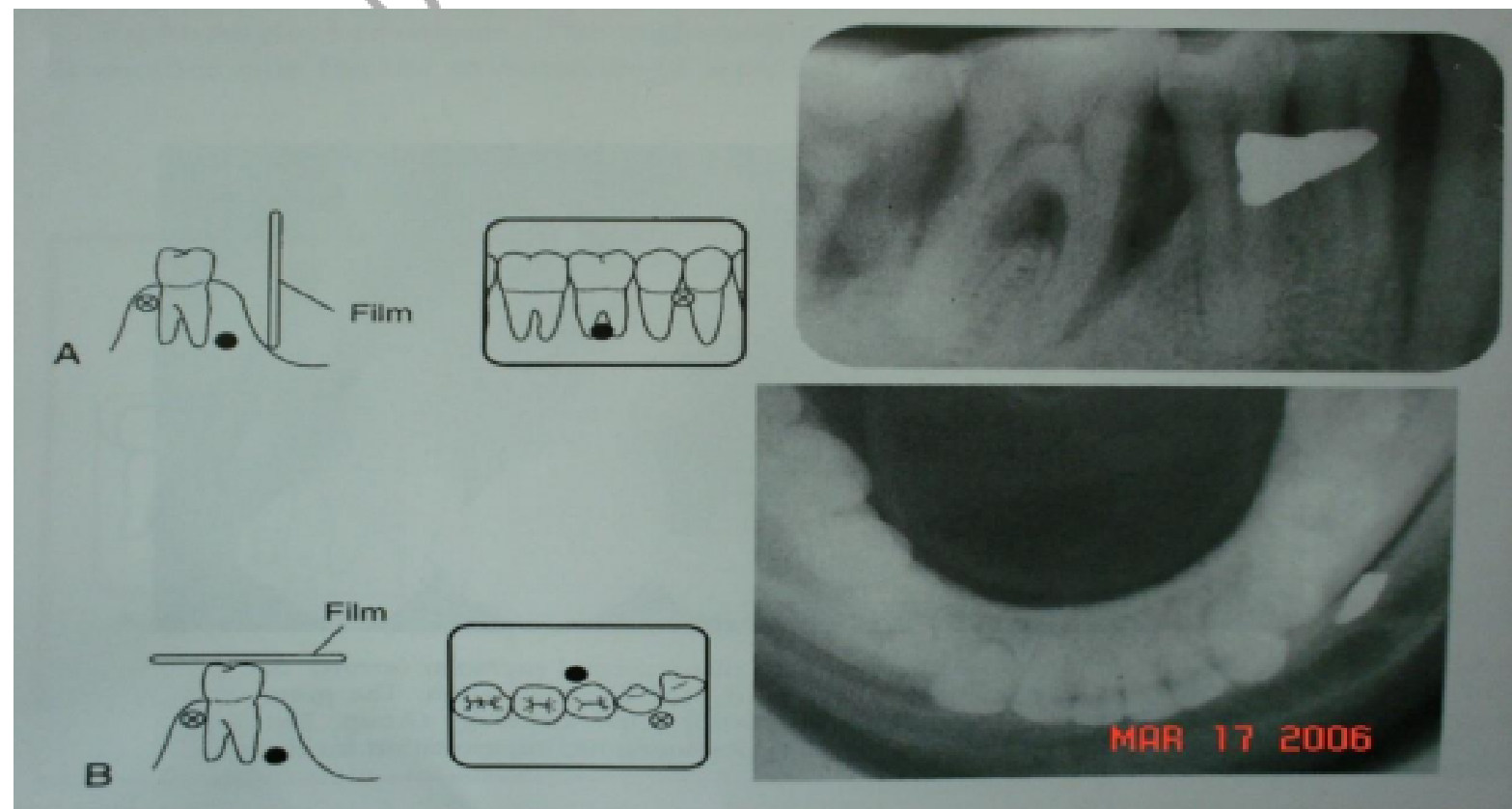
# Localization techniques

- ▢ A. Right angle technique.
  - ▢ B. Tube shift technique.
  - ▢ C. Stereo radiography (It is not widely used because it is time consuming and need a special viewing device).
  - ▢ D. Use of radio-opaque media
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# Right angle procedure:

- It involves the use of at least 2 films taken at right angles to each others.

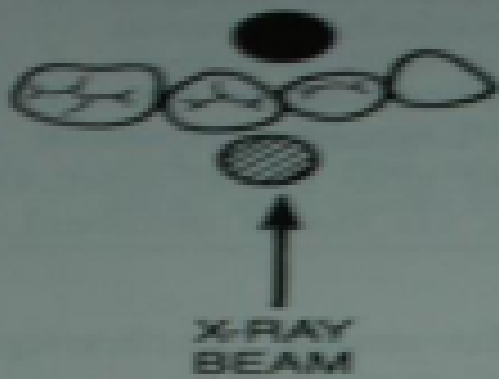


# Tube shift technique:Clark's rule

▮ The area in question is **anesthetized**, a small hypodermic **needle is inserted in vertical position in mucobuccal fold near the object in question**, a radiograph is taken, then insert another film and second radiograph is taken with mesial shifted tube. **The 2 films are processed and compared**



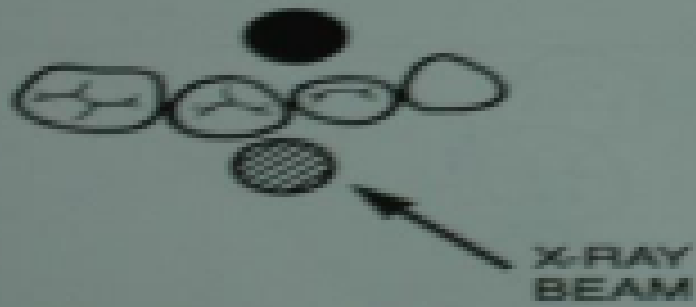
■ If the object in the second radiograph appears more mesialy, that means the object is located far lingualy or palately, while if it is more distally (in relation to the needle) it means it is buccaly positioned, and if it is not move it means that it is close to the needle. (SLOB) Same = Lingual , Opposite = Buccal



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# Use of radio-opaque media

▮: Barium sulfate, lipiodol and dionosil can be used to demonstrate cavernous areas with hard and soft structures. **After the injection of the radio-opaque media (mostly lipiodol) in cyst for example, film exposed, processed and viewed to see the extension of the cyst.** Radio-opaque media also used in sialogram to demonstrate the salivary glands and their duct.

